

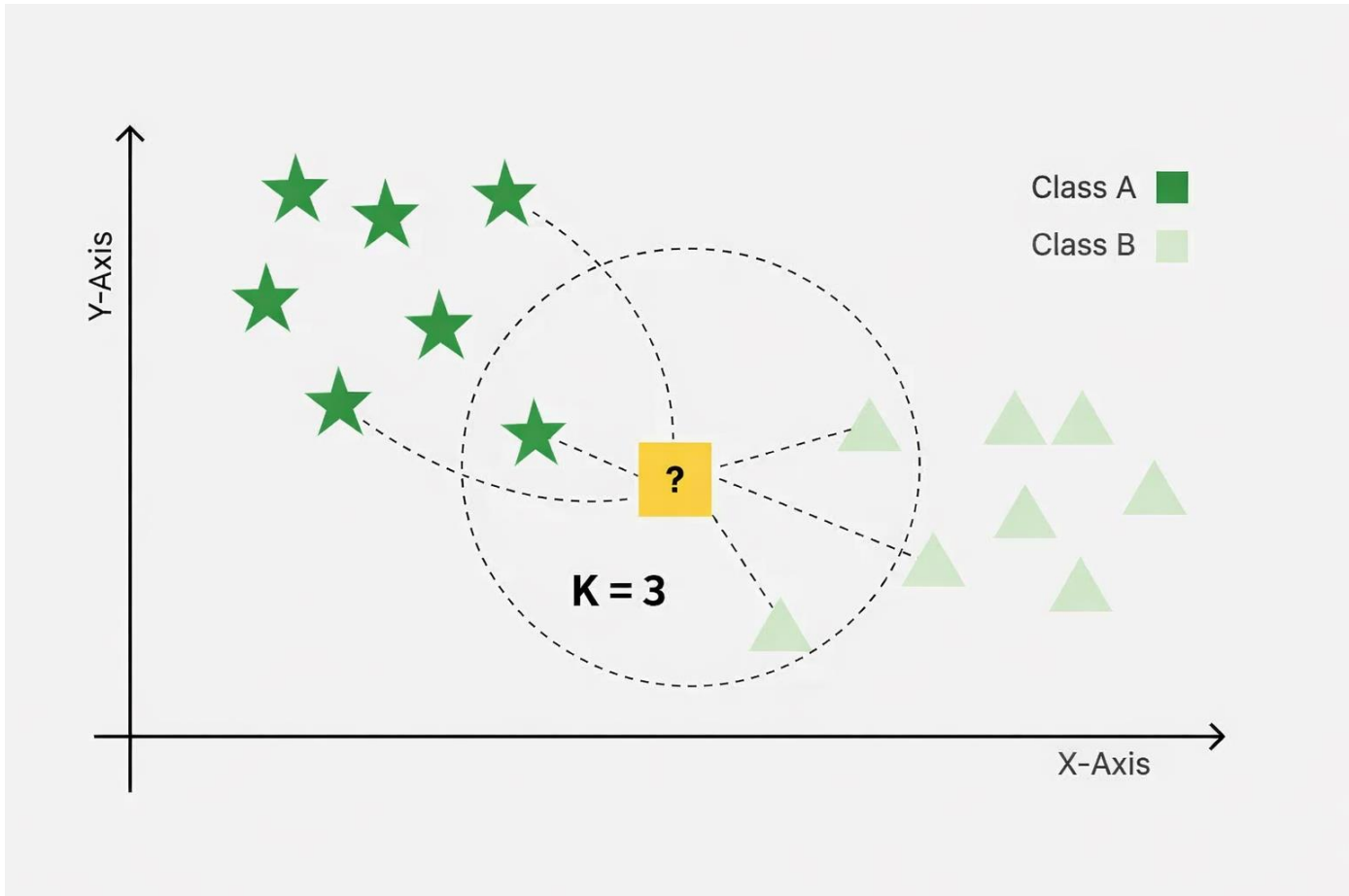
Module 7: Classification

Part 4: KNN

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K-nearest neighbor



K-nearest neighbor

- Takes a completely different approach than the other classifiers seen so far. *no equation, data decides the shape*
- KNN is completely non-parametric. Essentially, it makes **no underlying assumption** about the data or the shape of the decision boundary.
- We can expect it to dominate other classifiers when the decision boundary is **highly non-linear/irregular**.
- Drawback of KNN: you need a lot of data.

How it works ?

- Split data into training and test set.
- In order to classify an observation in the test set, I find the **K training observations** whose predictor values are closest to the test observation.
- Then the test observation is assigned to the class for which the **majority** of the K nearest training observations belong.

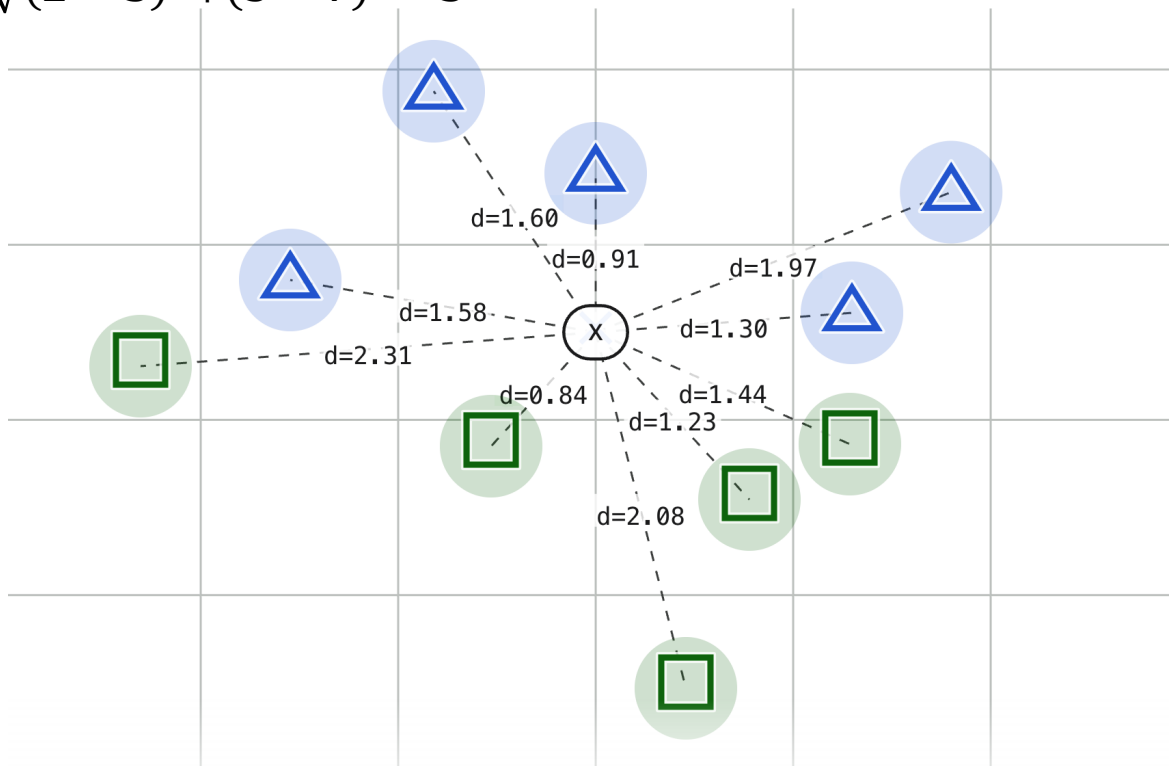
In-class activity

Step 1: Calculate distances

Euclidean distance:

Point A(2,3), Point B(5,7)

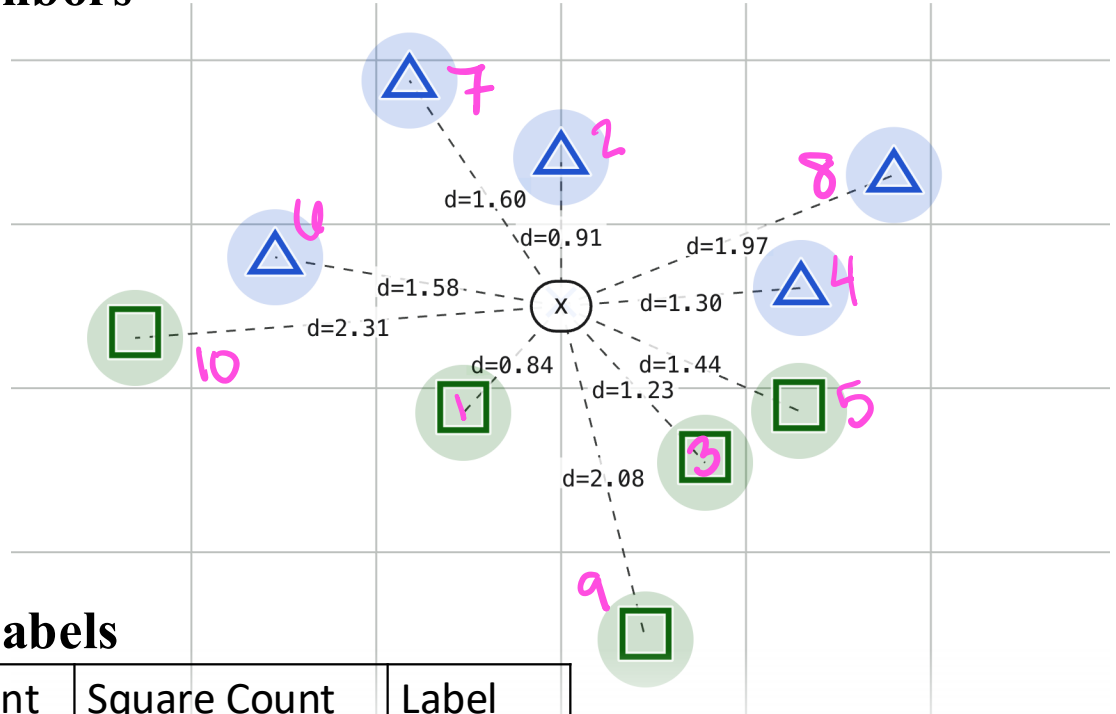
$$d = \sqrt{(2 - 5)^2 + (3 - 7)^2} = 5$$



In-class activity

Step 2: Find neighbors

1 st NN	0.84
2 nd NN	0.91
3 rd NN	1.23
4 th NN	1.30
5 th NN	1.44
6 th NN	1.58
7 th NN	1.60
....	



Step 3: Vote on labels

	Triangle Count	Square Count	Label
K = 1	0	1	□
K = 3	1	2	□
K = 5	2	3	□
K = 7	4	3	△

How can we handle ties for even k ?

↳ use odd value of k

↳ remove the furthest neighbor from consideration

↳ use a different method to calculate the distance

KNN Demo

1. <https://www.luseratech.com/knn-visual>
2. <http://vision.stanford.edu/teaching/cs231n-demos/knn/>
 - Drag one point near the boundary. What happens to the decision boundary for different k ?

How do we choose K ?

① test

training



②

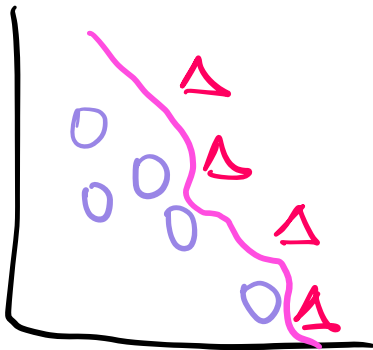
CV w/ $k=1, k=3, \dots, k=n$

③ Choose k w/ lowest

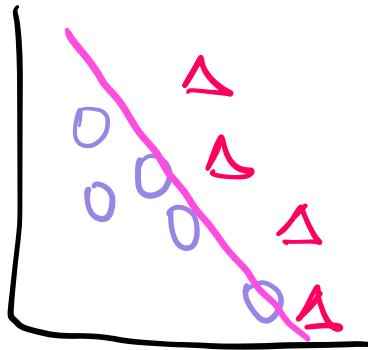
Choice of K affect bias-variance decomposition

Small $k \rightarrow$ "wiggly" (more flexible) $\rightarrow$ low bias,
high variance

large $k \rightarrow$ more smooth $\rightarrow$ high bias,
low variance



$k=1$



$k=7$

Definition of 'interest'

When the decision boundary is highly non-linear, KNN most closely mimics Bayes decision boundary.

How do we choose K ?

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Implementation

See R script: knn.R